FANUC America Corporation SYSTEM R-30*i*B/SYSTEM R-30*i*B Mate/SYSTEM R-30*i*B Plus/R-30*i*B Mate Plus/R-30*i*B Compact Plus System Variable Addendum

MARXUSVAR06181E REV A

To be used with your application-specific *Setup and Operations Manual*.

FANUC America Corporation
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About This Manual

If you have a controller labeled R-J3*i*C, you should read R-J3*i*C as R-30*i*A.

You should also read the *iRVision Visual Tracking Start-up Guidance manual* for a deeper understanding of PickTool. PickTool is a value-added application built on top of two key components — *i*RVision Visual Tracking and Line Tracking. This manual contains some descriptions that are taken from the *iRVision Visual Tracking Manual.*

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Warning

This equipment generates, uses, and can radiate radio frequency energy and if not installed and used in accordance with the instruction manual, may cause interference to radio communications. As temporarily permitted by regulation, it has not been tested for compliance with the limits for Class A computing devices pursuant to subpart J of Part 15 of FCC Rules, which are designed to provide reasonable protection against such interference. Operation of the equipment in a residential area is likely to cause interference, in which case the user, at his own expense, will be required to take whatever measure may be required to correct the interference.

FANUC America conducts courses on its systems and products on a regularly scheduled basis at its headquarters in Rochester Hills, Michigan. For additional information contact

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For customer assistance, including Technical Support, Service, Parts & Part Repair, and Marketing Requests, contact the Customer Resource Center, 24 hours a day, at 888-FANUC-US (888-326-8287).

Send your comments and suggestions about this manual to: product.documentation@fanucamerica.com

Mate ™, SR ShotTool ™, SureWeld™, SYSTEM R-J2 Controller R-J3*i* B Controller™, SYSTEM R-J3*i* C Controller™, SYSTEM R-30*i* A Controller™, SYSTEM R-30*i* A Mate Controller™, SYSTEM R-30*i* B Controller™, SYSTEM R-30*i* B Mate Controller™, SYSTEM R-30*i* B Plus Controller™, SYSTEM R-30*i* B Mate Plus Controller™, TCP Mate ™, TorchMate ™, TripleARM ™, TurboMove™, visLOC ™, visTRAC ™, WebServer ™, WebTP ™, and YagTool™.

Patents

One or more of the following U.S. patents might be related to the FANUC America products described in this manual.

FRA Patent List

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VersaBell, ServoBell and SpeedDock Patents Pending.

Conventions

This manual includes information essential to the safety of personnel, equipment, software, and data. This information is indicated by headings and boxes in the text.

> **Warning**
>
> **Information appearing under WARNING concerns the protection of personnel. It is boxed and in bold type to set it apart from other text.**

> **Caution**
>
> Information appearing under **CAUTION** concerns the protection of equipment, software, and data. It is boxed to set it apart from other text.

Note Information appearing next to **NOTE** concerns related information or useful hints.

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Safety

FANUC America Corporation is not and does not represent itself as an expert in safety systems, safety equipment, or the specific safety aspects of your company and/or its work force. It is the responsibility of the owner, employer, or user to take all necessary steps to guarantee the safety of all personnel in the workplace.

The appropriate level of safety for your application and installation can best be determined by safety system professionals. FANUC America Corporation therefore, recommends that each customer consult with such professionals in order to provide a workplace that allows for the safe application, use, and operation of FANUC America Corporation systems.

According to the industry standard ANSI/RIA R15-06, the owner or user is advised to consult the standards to ensure compliance with its requests for Robotics System design, usability, operation, maintenance, and service. Additionally, as the owner, employer, or user of a robotic system, it is your responsibility to arrange for the training of the operator of a robot system to recognize and respond to known hazards associated with your robotic system and to be aware of the recommended operating procedures for your particular application and robot installation.

Ensure that the robot being used is appropriate for the application. Robots used in classified (hazardous) locations must be certified for this use.

FANUC America Corporation therefore, recommends that all personnel who intend to operate, program, repair, or otherwise use the robotics system be trained in an approved FANUC America Corporation training course and become familiar with the proper operation of the system. Persons responsible for programming the system-including the design, implementation, and debugging of application programs-must be familiar with the recommended programming procedures for your application and robot installation.

The following guidelines are provided to emphasize the importance of safety in the workplace.

CONSIDERING SAFETY FOR YOUR ROBOT INSTALLATION

Safety is essential whenever robots are used. Keep in mind the following factors with regard to safety:

- The safety of people and equipment
- Use of safety enhancing devices
- Techniques for safe teaching and manual operation of the robot(s)
- Techniques for safe automatic operation of the robot(s)
- Regular scheduled inspection of the robot and workcell
- Proper maintenance of the robot

Keeping People Safe

The safety of people is always of primary importance in any situation. When applying safety measures to your robotic system, consider the following:

- External devices
- Robot(s)
- Tooling
- Workpiece

Using Safety Enhancing Devices

Always give appropriate attention to the work area that surrounds the robot. The safety of the work area can be enhanced by the installation of some or all of the following devices:

- Safety fences, barriers, or chains
- Light curtains
- Interlocks
- Pressure mats
- Floor markings
- Warning lights
- Mechanical stops
- EMERGENCY STOP buttons
- DEADMAN switches

Setting Up a Safe Workcell

A safe workcell is essential to protect people and equipment. Observe the following guidelines to ensure that the workcell is set up safely. These suggestions are intended to supplement and **not** replace existing federal, state, and local laws, regulations, and guidelines that pertain to safety.

- Sponsor your personnel for training in approved FANUC America Corporation training course(s) related to your application. Never permit untrained personnel to operate the robots.
- Install a lockout device that uses an access code to prevent unauthorized persons from operating the robot.
- Use anti-tie-down logic to prevent the operator from bypassing safety measures.
- Arrange the workcell so the operator faces the workcell and can see what is going on inside the cell.

- Clearly identify the work envelope of each robot in the system with floor markings, signs, and special barriers. The work envelope is the area defined by the maximum motion range of the robot, including any tooling attached to the wrist flange that extend this range.
- Position all controllers outside the robot work envelope.
- Never rely on software or firmware based controllers as the primary safety element unless they comply with applicable current robot safety standards.
- Mount an adequate number of EMERGENCY STOP buttons or switches within easy reach of the operator and at critical points inside and around the outside of the workcell.
- Install flashing lights and/or audible warning devices that activate whenever the robot is operating, that is, whenever power is applied to the servo drive system. Audible warning devices shall exceed the ambient noise level at the end-use application.
- Wherever possible, install safety fences to protect against unauthorized entry by personnel into the work envelope.
- Install special guarding that prevents the operator from reaching into restricted areas of the work envelope.
- Use interlocks.
- Use presence or proximity sensing devices such as light curtains, mats, and capacitance and vision systems to enhance safety.
- Periodically check the safety joints or safety clutches that can be optionally installed between the robot wrist flange and tooling. If the tooling strikes an object, these devices dislodge, remove power from the system, and help to minimize damage to the tooling and robot.
- Make sure all external devices are properly filtered, grounded, shielded, and suppressed to prevent hazardous motion due to the effects of electro-magnetic interference (EMI), radio frequency interference (RFI), and electro-static discharge (ESD).
- Make provisions for power lockout/tagout at the controller.
- Eliminate *pinch points* . Pinch points are areas where personnel could get trapped between a moving robot and other equipment.
- Provide enough room inside the workcell to permit personnel to teach the robot and perform maintenance safely.
- Program the robot to load and unload material safely.
- If high voltage electrostatics are present, be sure to provide appropriate interlocks, warning, and beacons.
- If materials are being applied at dangerously high pressure, provide electrical interlocks for lockout of material flow and pressure.

Staying Safe While Teaching or Manually Operating the Robot

Advise all personnel who must teach the robot or otherwise manually operate the robot to observe the following rules:

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Know whether or not you are using an intrinsically safe teach pendant if you are working in a hazardous environment.
- Before teaching, visually inspect the robot and *work envelope* to make sure that no potentially hazardous conditions exist. The work envelope is the area defined by the maximum motion range of the robot. These include tooling attached to the wrist flange that extends this range.
- The area near the robot must be clean and free of oil, water, or debris. Immediately report unsafe working conditions to the supervisor or safety department.
- FANUC America Corporation recommends that no one enter the work envelope of a robot that is on, except for robot teaching operations. However, if you must enter the work envelope, be sure all safeguards are in place, check the teach pendant DEADMAN switch for proper operation, and place the robot in teach mode. Take the teach pendant with you, turn it on, and be prepared to release the DEADMAN switch. Only the person with the teach pendant should be in the work envelope.

Warning

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

- Know the path that can be used to escape from a moving robot; make sure the escape path is never blocked.
- Isolate the robot from all remote control signals that can cause motion while data is being taught.
- Test any program being run for the first time in the following manner:

Warning

Stay outside the robot work envelope whenever a program is being run. Failure to do so can result in injury.

— Using a low motion speed, single step the program for at least one full cycle.

— Using a low motion speed, test run the program continuously for at least one full cycle.

— Using the programmed speed, test run the program continuously for at least one full cycle.

- Make sure all personnel are outside the work envelope before running production.

Staying Safe During Automatic Operation

Advise all personnel who operate the robot during production to observe the following rules:

- Make sure all safety provisions are present and active.
- Know the entire workcell area. The workcell includes the robot and its work envelope, plus the area occupied by all external devices and other equipment with which the robot interacts.
- Understand the complete task the robot is programmed to perform before initiating automatic operation.
- Make sure all personnel are outside the work envelope before operating the robot.
- Never enter or allow others to enter the work envelope during automatic operation of the robot.
- Know the location and status of all switches, sensors, and control signals that could cause the robot to move.
- Know where the EMERGENCY STOP buttons are located on both the robot control and external control devices. Be prepared to press these buttons in an emergency.
- Never assume that a program is complete if the robot is not moving. The robot could be waiting for an input signal that will permit it to continue activity.
- If the robot is running in a pattern, do not assume it will continue to run in the same pattern.
- Never try to stop the robot, or break its motion, with your body. The only way to stop robot motion immediately is to press an EMERGENCY STOP button located on the controller panel, teach pendant, or emergency stop stations around the workcell.

Staying Safe During Inspection

When inspecting the robot, be sure to

- Turn off power at the controller.
- Lock out and tag out the power source at the controller according to the policies of your plant.
- Turn off the compressed air source and relieve the air pressure.
- If robot motion is not needed for inspecting the electrical circuits, press the EMERGENCY STOP button on the operator panel.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- If power is needed to check the robot motion or electrical circuits, be prepared to press the EMERGENCY STOP button, in an emergency.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

Staying Safe During Maintenance

When performing maintenance on your robot system, observe the following rules:

- Never enter the work envelope while the robot or a program is in operation.
- Before entering the work envelope, visually inspect the workcell to make sure no potentially hazardous conditions exist.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Consider all or any overlapping work envelopes of adjoining robots when standing in a work envelope.
- Test the teach pendant for proper operation before entering the work envelope.
- If it is necessary for you to enter the robot work envelope while power is turned on, you must be sure that you are in control of the robot. Be sure to take the teach pendant with you, press the DEADMAN switch, and turn the teach pendant on. Be prepared to release the DEADMAN switch to turn off servo power to the robot immediately.
- Whenever possible, perform maintenance with the power turned off. Before you open the controller front panel or enter the work envelope, turn off and lock out the 3-phase power source at the controller.
- Be aware that an applicator bell cup can continue to spin at a very high speed even if the robot is idle. Use protective gloves or disable bearing air and turbine air before servicing these items.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

Warning

Lethal voltage is present in the controller WHENEVER IT IS CONNECTED to a power source. Be extremely careful to avoid electrical shock. HIGH VOLTAGE IS PRESENT at the input side whenever the controller is connected to a power source. Turning the disconnect or circuit breaker to the OFF position removes power from the output side of the device only.

- Release or block all stored energy. Before working on the pneumatic system, shut off the system air supply and purge the air lines.
- Isolate the robot from all remote control signals. If maintenance must be done when the power is on, make sure the person inside the work envelope has sole control of the robot. The teach pendant must be held by this person.
- Make sure personnel cannot get trapped between the moving robot and other equipment. Know the path that can be used to escape from a moving robot. Make sure the escape route is never blocked.

- Use blocks, mechanical stops, and pins to prevent hazardous movement by the robot. Make sure that such devices do not create pinch points that could trap personnel.

> **Warning**
>
> **Do not try to remove any mechanical component from the robot before thoroughly reading and understanding the procedures in the appropriate manual. Doing so can result in serious personal injury and component destruction.**

- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.
- When replacing or installing components, make sure dirt and debris do not enter the system.
- Use only specified parts for replacement. To avoid fires and damage to parts in the controller, never use nonspecified fuses.
- Before restarting a robot, make sure no one is inside the work envelope; be sure that the robot and all external devices are operating normally.

KEEPING MACHINE TOOLS AND EXTERNAL DEVICES SAFE

Certain programming and mechanical measures are useful in keeping the machine tools and other external devices safe. Some of these measures are outlined below. Make sure you know all associated measures for safe use of such devices.

Programming Safety Precautions

Implement the following programming safety measures to prevent damage to machine tools and other external devices.

- Back-check limit switches in the workcell to make sure they do not fail.
- Implement “failure routines” in programs that will provide appropriate robot actions if an external device or another robot in the workcell fails.
- Use *handshaking* protocol to synchronize robot and external device operations.
- Program the robot to check the condition of all external devices during an operating cycle.

Mechanical Safety Precautions

Implement the following mechanical safety measures to prevent damage to machine tools and other external devices.

- Make sure the workcell is clean and free of oil, water, and debris.
- Use DCS (Dual Check Safety), software limits, limit switches, and mechanical hardstops to prevent undesired movement of the robot into the work area of machine tools and external devices.

KEEPING THE ROBOT SAFE

Observe the following operating and programming guidelines to prevent damage to the robot.

Operating Safety Precautions

The following measures are designed to prevent damage to the robot during operation.

- Use a low override speed to increase your control over the robot when jogging the robot.
- Visualize the movement the robot will make before you press the jog keys on the teach pendant.
- Make sure the work envelope is clean and free of oil, water, or debris.
- Use circuit breakers to guard against electrical overload.

Programming Safety Precautions

The following safety measures are designed to prevent damage to the robot during programming:

- Establish *interference zones* to prevent collisions when two or more robots share a work area.
- Make sure that the program ends with the robot near or at the home position.
- Be aware of signals or other operations that could trigger operation of tooling resulting in personal injury or equipment damage.
- In dispensing applications, be aware of all safety guidelines with respect to the dispensing materials.

  Note Any deviation from the methods and safety practices described in this manual must conform to the approved standards of your company. If you have questions, see your supervisor.

ADDITIONAL SAFETY CONSIDERATIONS FOR PAINT ROBOT INSTALLATIONS

Process technicians are sometimes required to enter the paint booth, for example, during daily or routine calibration or while teaching new paths to a robot. Maintenance personal also must work inside the paint booth periodically.

Whenever personnel are working inside the paint booth, ventilation equipment must be used. Instruction on the proper use of ventilating equipment usually is provided by the paint shop supervisor.

Although paint booth hazards have been minimized, potential dangers still exist. Therefore, today's highly automated paint booth requires that process and maintenance personnel have full awareness of the system and its capabilities. They must understand the interaction that occurs between the vehicle moving along the conveyor and the robot(s), hood/deck and door opening devices, and high-voltage electrostatic tools.

Caution

Ensure that all ground cables remain connected. Never operate the paint robot with ground provisions disconnected. Otherwise, you could injure personnel or damage equipment.

Paint robots are operated in three modes:

- Teach or manual mode
- Automatic mode, including automatic and exercise operation
- Diagnostic mode

During both teach and automatic modes, the robots in the paint booth will follow a predetermined pattern of movements. In teach mode, the process technician teaches (programs) paint paths using the teach pendant.

In automatic mode, robot operation is initiated at the System Operator Console (SOC) or Manual Control Panel (MCP), if available, and can be monitored from outside the paint booth. All personnel must remain outside of the booth or in a designated safe area within the booth whenever automatic mode is initiated at the SOC or MCP.

In automatic mode, the robots will execute the path movements they were taught during teach mode, but generally at production speeds.

When process and maintenance personnel run diagnostic routines that require them to remain in the paint booth, they must stay in a designated safe area.

Paint System Safety Features

Process technicians and maintenance personnel must become totally familiar with the equipment and its capabilities. To minimize the risk of injury when working near robots and related equipment, personnel must comply strictly with the procedures in the manuals.

This section provides information about the safety features that are included in the paint system and also explains the way the robot interacts with other equipment in the system.

The paint system includes the following safety features:

- Most paint booths have red warning beacons that illuminate when the robots are armed and ready to paint. Your booth might have other kinds of indicators. Learn what these are.
- Some paint booths have a blue beacon that, when illuminated, indicates that the electrostatic devices are enabled. Your booth might have other kinds of indicators. Learn what these are.
- EMERGENCY STOP buttons are located on the robot controller and teach pendant. Become familiar with the locations of all E-STOP buttons.
- An intrinsically safe teach pendant is used when teaching in hazardous paint atmospheres.
- A DEADMAN switch is located on each teach pendant. When this switch is held in, and the teach pendant is on, power is applied to the robot servo system. If the engaged DEADMAN switch is released during robot operation, power is removed from the servo system, all axis brakes are applied, and the robot comes to an EMERGENCY STOP. Safety interlocks within the system might also E-STOP other robots.

> **Warning**
>
> **An EMERGENCY STOP will occur if the DEADMAN switch is released on a bypassed robot.**

- Overtravel by robot axes is prevented by software limits. All of the major and minor axes are governed by software limits. DCS (Dual Check Safety), limit switches and hardstops also limit travel by the major axes.
- EMERGENCY STOP limit switches and photoelectric eyes might be part of your system. Limit switches, located on the entrance/exit doors of each booth, will EMERGENCY STOP all equipment in the booth if a door is opened while the system is operating in automatic or manual mode. For some systems, signals to these switches are inactive when the switch on the SOC is in teach mode.When present, photoelectric eyes are sometimes used to monitor unauthorized intrusion through the entrance/exit silhouette openings.
- System status is monitored by computer. Severe conditions result in automatic system shutdown.

Staying Safe While Operating the Paint Robot

When you work in or near the paint booth, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.

> **Warning**
>
> **Observe all safety rules and guidelines to avoid injury.**

> **Warning**
>
> **Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.**

> **Warning**
>
> **Enclosures shall not be opened unless the area is known to be nonhazardous or all power has been removed from devices within the enclosure. Power shall not be restored after the enclosure has been opened until all combustible dusts have been removed from the interior of the enclosure and the enclosure purged. Refer to the Purge chapter for the required purge time.**

- Know the work area of the entire paint station (workcell).
- Know the work envelope of the robot and hood/deck and door opening devices.
- Be aware of overlapping work envelopes of adjacent robots.
- Know where all red, mushroom-shaped EMERGENCY STOP buttons are located.
- Know the location and status of all switches, sensors, and/or control signals that might cause the robot, conveyor, and opening devices to move.
- Make sure that the work area near the robot is clean and free of water, oil, and debris. Report unsafe conditions to your supervisor.
- Become familiar with the complete task the robot will perform BEFORE starting automatic mode.
- Make sure all personnel are outside the paint booth before you turn on power to the robot servo system.
- Never enter the work envelope or paint booth before you turn off power to the robot servo system.
- Never enter the work envelope during automatic operation unless a safe area has been designated.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.

- Remove all metallic objects, such as rings, watches, and belts, before entering a booth when the electrostatic devices are enabled.
- Stay out of areas where you might get trapped between a moving robot, conveyor, or opening device and another object.
- Be aware of signals and/or operations that could result in the triggering of guns or bells.
- Be aware of all safety precautions when dispensing of paint is required.
- Follow the procedures described in this manual.

Special Precautions for Combustible Dusts (powder paint)

When the robot is used in a location where combustible dusts are found, such as the application of powder paint, the following special precautions are required to insure that there are no combustible dusts inside the robot.

- Purge maintenance air should be maintained at all times, even when the robot power is off. This will insure that dust can not enter the robot.
- A purge cycle will not remove accumulated dusts. Therefore, if the robot is exposed to dust when maintenance air is not present, it will be necessary to remove the covers and clean out any accumulated dust. Do not energize the robot until you have performed the following steps.

1. Before covers are removed, the exterior of the robot should be cleaned to remove accumulated dust.
2. When cleaning and removing accumulated dust, either on the outside or inside of the robot, be sure to use methods appropriate for the type of dust that exists. Usually lint free rags dampened with water are acceptable. Do not use a vacuum cleaner to remove dust as it can generate static electricity and cause an explosion unless special precautions are taken.
3. Thoroughly clean the interior of the robot with a lint free rag to remove any accumulated dust.
4. When the dust has been removed, the covers must be replaced immediately.
5. Immediately after the covers are replaced, run a complete purge cycle. The robot can now be energized.

Staying Safe While Operating Paint Application Equipment

When you work with paint application equipment, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.

> **Warning**
>
> **When working with electrostatic paint equipment, follow all national and local codes as well as all safety guidelines within your organization. Also reference the following standards: *NFPA 33 Standards for Spray Application Using Flammable or Combustible Materials* , and *NFPA 70 National Electrical Code* .**

- **Grounding**: All electrically conductive objects in the spray area must be grounded. This includes the spray booth, robots, conveyors, workstations, part carriers, hooks, paint pressure pots, as well as solvent containers. Grounding is defined as the object or objects shall be electrically connected to ground with a resistance of not more than 1 megohms.
- **High Voltage**: High voltage should only be on during actual spray operations. Voltage should be off when the painting process is completed. Never leave high voltage on during a cap cleaning process.
- Avoid any accumulation of combustible vapors or coating matter.
- Follow all manufacturer recommended cleaning procedures.
- Make sure all interlocks are operational.
- No smoking.
- Post all warning signs regarding the electrostatic equipment and operation of electrostatic equipment according to NFPA 33 Standard for Spray Application Using Flammable or Combustible Material.
- Disable all air and paint pressure to bell.
- Verify that the lines are not under pressure.

Staying Safe During Maintenance

When you perform maintenance on the painter system, observe the following rules, and all other maintenance safety rules that apply to all robot installations. Only qualified, trained service or maintenance personnel should perform repair work on a robot.

- Paint robots operate in a potentially explosive environment. Use caution when working with electric tools.
- When a maintenance technician is repairing or adjusting a robot, the work area is under the control of that technician. All personnel not participating in the maintenance must stay out of the area.
- For some maintenance procedures, station a second person at the control panel within reach of the EMERGENCY STOP button. This person must understand the robot and associated potential hazards.
- Be sure all covers and inspection plates are in good repair and in place.
- Always return the robot to the ``home'' position before you disarm it.

- Never use machine power to aid in removing any component from the robot.
- During robot operations, be aware of the robot's movements. Excess vibration, unusual sounds, and so forth, can alert you to potential problems.
- Whenever possible, turn off the main electrical disconnect before you clean the robot.
- When using vinyl resin observe the following:
  - — Wear eye protection and protective gloves during application and removal
  - — Adequate ventilation is required. Overexposure could cause drowsiness or skin and eye irritation.
  - — If there is contact with the skin, wash with water.
  - — Follow the Original Equipment Manufacturer's Material Safety Data Sheets.
- When using paint remover observe the following:
  - — Eye protection, protective rubber gloves, boots, and apron are required during booth cleaning.
  - — Adequate ventilation is required. Overexposure could cause drowsiness.
  - — If there is contact with the skin or eyes, rinse with water for at least 15 minutes. Then, seek medical attention as soon as possible.
  - — Follow the Original Equipment Manufacturer's Material Safety Data Sheets.

Appendix A

SYSTEM VARIABLES

Contents

This sppendix describes the names, functions, standard settings, and valid ranges of system variables.

A.1 FORMAT OF A SYSTEM VARIABLE

Figure A–1. System Variable Format

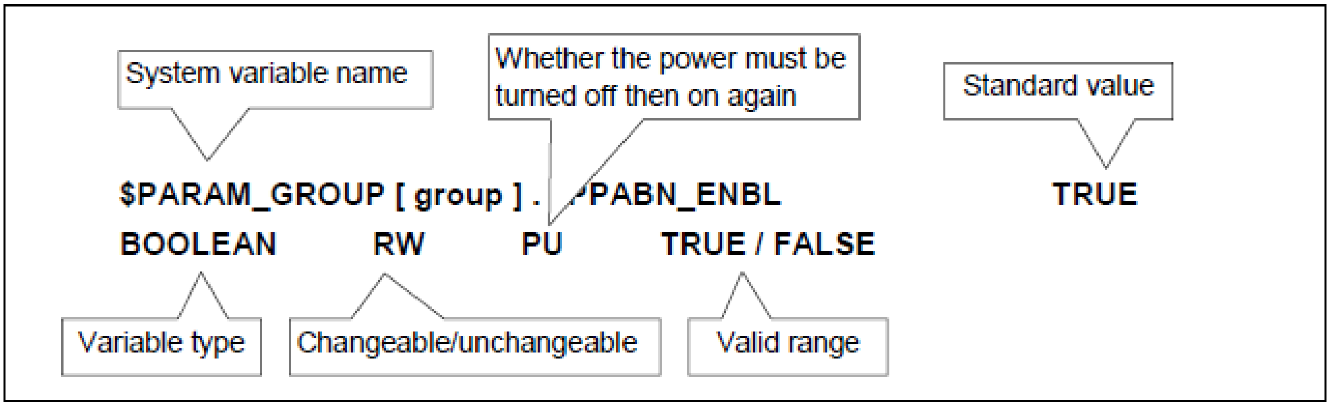

Table A–1. Format of a system variable

System variable name	
Standard value	* Intrinsic value for each model
Variable type	BOOLEAN True/false type (TRUE/FALSE)
	BYTE Integer (0 to 255)
	SHORT Integer (-32768 to 32767)
	INTEGER Integer (-1000000 to 1000000)
	REAL Real number (-10000000000 to 1000000000)
	CHAR Character string (“abcdefg”)
	XYZWPR Cartesian coordinates
Changeable/unchangeable	RW Changeable
	RO Unchangeable
Whether the power must be turned off then on again	PU Indicates that the power must be turned on again.
Valid range (unit)	

Procedure A-1 Setting a System Variable

1. Press the [MENU] key.
2. Select 0 (NEXT), then select 6 (SYSTEM).
3. Press the F1, [TYPE] key.
4. Select Variables. Then, the system variable screen will be displayed.

```
SYSTEM Variables
                                              1/638
       1 $AAVM                   AAVM_T
       2 $ABSPOS_GRP             ABSPOS_GRP_T
       3 $ACC_MAXLMT             150
       4 $ACC_MINLMT             0
       5 $ACC_PRE_EXE            0
       6 $ALM_IF                 ALM_IF_T
       7 $ANGTOL                 [9] of REAL
       8 $APPLICATION            [9] of STRING[21]
       9 $AP_ACTIVE              6
      10 $AP_AUTOMODE            FALSE
      11 $AP_CHGAPONL            TRUE
 [ TYPE ] DETAIL
```

5. To change the value of a system variable, move the cursor to a desired item, enter a new value, then press the [ENTER] key or select a desired item by pressing the corresponding function key.
6. When a system variable contains multiple system variables, move the cursor to a desired item and press the [ENTER] key. Then, the low-order system variables are displayed.

```
SYSTEM Variables
                                           329/638
     327 $PADJ_SCHNUM            10
     328 $PARAM2_GRP             MRR2_GRP_T
     329 $PARAM_GROUP            MRR_GRP_T
     330 $PARAM_MENU             [21] of STRING[21]
     331 $PASSNAME               [10] of PASSNAME_T
     332 $PASSSUPER              PASSNAME_T
     333 $PASSWORD               PASSWORD_T
     334 $PAUSE_PROG             *uninit*
     335 $PCCRT                  0
     336 $PCCRT_HOST
'PCCRT'
     337 $PCTP                   0
 [ TYPE ] DETAIL
```

```
SYSTEM Variables
     $PARAM_GROUP[1]                            1/236
        1 $BELT_ENABLE          FALSE
        2 $CART_ACCEL1          800
        3 $CART_ACCEL2          400
        4 $CIRC_RATE            1
        5 $CONTAXISNUM          0
        6 $EXP_ENBL             FALSE
        7 $JOINT_RATE           1
        8 $LINEAR_RATE          1
        9 $PATH_ACCEL1          800
       10 $PATH_ACCEL2          400
       11 $PROCESS_SPD          2000.000
 [ TYPE ]   TRUE   FALSE
```

7. After changing the setting of the system variable for which PU is specified, cycle power of the controller. (PU is specified for all $PARAM_GROUP system variables.)

   Note The setting of a system variable for which RO (unchangeable) is specified cannot be changed.

A.2 SYSTEM VARIABLES

A.2.1 Power Fail Recovery

$SEMIPOWERFL FALSE

BOOLEAN RW TRUE/FALSE

Table A–2. Power Fail Recovery — $SEMIPOWERFL

[Function]	Enables or disables the hot start
[Description]	Specifies whether to perform a hot start when power is recovered. After a hot start, the robot is restored near the status immediately before a power failure. TRUE: Performs a hot start after power recovery. FALSE: Does not perform a hot start. Instead, performs a cold start.

A.2.2 Brake Control

$PARAM_GROUP[group] . $SV_OFF_ENB[1] *

$PARAM_GROUP[group] . $SV_OFF_ENB[2] *

$PARAM_GROUP[group] . $SV_OFF_ENB[3] *

$PARAM_GROUP[group] . $SV_OFF_ENB[4] *

$PARAM_GROUP[group] . $SV_OFF_ENB[5] *

$PARAM_GROUP[group] . $SV_OFF_ENB[6] *

$PARAM_GROUP[group] . $SV_OFF_ENB[7] *

$PARAM_GROUP[group] . $SV_OFF_ENB[8] *

$PARAM_GROUP[group] . $SV_OFF_ENB[9] *

BOOLEAN RW PU TRUE / FALSE

Table A–3. Brake Control — $PARAM_GROUP[group].$SV_OFF_ENB

[Function]	Enables or disables the brake control function
[Description]	Specifies how the brake is applied when the axis doesn't move for a given length of time. However the brake control function is disabled for all axes on the same brake release DO if there is at least one axis that the brake control is disabled on the DO. TRUE: It applies the brake when the axis doesn't move for a given length of time and it releases the brake when the axis starts to move. FALSE: It never applies the brake even if the axis doesn't move for a given length of time.

$PARAM_GROUP[group] . $SV_OFF_TIME[1] *

$PARAM_GROUP[group] . $SV_OFF_TIME[2] *

$PARAM_GROUP[group] . $SV_OFF_TIME[3] *

$PARAM_GROUP[group] . $SV_OFF_TIME[4] *

$PARAM_GROUP[group] . $SV_OFF_TIME[5] *

$PARAM_GROUP[group] . $SV_OFF_TIME[6] *

$PARAM_GROUP[group] . $SV_OFF_TIME[7] *

$PARAM_GROUP[group] . $SV_OFF_TIME[8] *

$PARAM_GROUP[group] . $SV_OFF_TIME[9] *

INTEGER RW PU 0 to 100000000 (msec)

Table A–4. Brake Control — $PARAM_GROUP[group].$SV_OFF_TIME

[Function]	Time that elapses before the brake control function performs
[Description]	Specifies the elapsed time [msec] before it applies the brake by the brake control function.

$PARAM_GROUP[group] . $SV_OFF_ALL TRUE

BOOLEAN RW PU TRUE / FALSE

Table A–5. Brake Control — $PARAM_GROUP[group].$SV_OFF_ALL

[Function]	Enables or disables the synchronous brake control for all axes
[Description]	Specifies how the brakes for all axes are applied at the same time. TRUE: It applies/releases brakes for all axes at same time, i.e. it does not put on all brakes till all axes finish to move and it puts off all brakes when one axis starts to move. FALSE: It applies/releases brakes independently. However brakes on the same brake release DO are not applied/released independently because brakes for all axes on the same brake release DO are controlled together.

A.2.3 Mastering

$MASTER_ENB 0

ULONG RW 1 / 0

Table A–6. Mastering — $MASTER_ENB

[Function]	Displays positioning screen
[Description]	When this variable is enabled, the positioning screen [6 SYSTEM, Master/Cal] is displayed on the teach pendant. 0: Positioning screen not displayed. 1: Positioning screen displayed.

$DMR_GRP[group]. $MASTER_DONE TRUE

BOOLEAN RW TRUE / FALSE

Table A–7. Mastering — $DMR_GRP[group].$MASTER_DONE

[Function]	Indicates if mastering is completed.
[Description]	Indicates if mastering has been completed.
[Setting]	On the positioning screen [6 SYSTEM, Master/Cal]

$DMR_GRP[group]. $MASTER_COUN[1] *

$DMR_GRP[group]. $MASTER_COUN[2] *

$DMR_GRP[group]. $MASTER_COUN[3] *

$DMR_GRP[group]. $MASTER_COUN[4] *

$DMR_GRP[group]. $MASTER_COUN[5] *

$DMR_GRP[group]. $MASTER_COUN[6] *

$DMR_GRP[group]. $MASTER_COUN[7] *

$DMR_GRP[group]. $MASTER_COUN[8] *

$DMR_GRP[group]. $MASTER_COUN[9] *

INTEGER RW 0 to 100000000 (pulse)

Table A–8. Mastering — $DMR_GRP[group].$MASTER_COUN

[Function]	Store mastering pulse counts
[Description]	Pulsecoder count at zero degree position is stored. This value is calculated from current count at mastering and current position.

$DMR_GRP[group]. $GRAV_MAST *

INTEGER RW -1 to 1

Table A–9. Mastering — $DMR_GRP[group].$GRAV_MAST

[Function]	Indicates if Mastering counts are taken with Gravity Compensation or not.
[Description]	Indicates whether mastering counts are derived from Mastering with Gravity Compensation enabled or not. For details, refer to “Gravity Compensation” in your application-specific Setup and Operations Manual. 1 Current mastering counts are derived from Mastering with Gravity Compensation. 0 Current mastering counts are derived from Mastering without Gravity Compensation. -1 Unknown (not set).

$PARAM_GROUP[group]. $MASTER_POS[1] *

$PARAM_GROUP[group]. $MASTER_POS[2] *

$PARAM_GROUP[group]. $MASTER_POS[3] *

$PARAM_GROUP[group]. $MASTER_POS[4] *

$PARAM_GROUP[group]. $MASTER_POS[5] *

$PARAM_GROUP[group]. $MASTER_POS[6] *

$PARAM_GROUP[group]. $MASTER_POS[7] *

$PARAM_GROUP[group]. $MASTER_POS[8] *

$PARAM_GROUP[group]. $MASTER_POS[9] *

REAL RW PU -100000 to 100000 (deg) *

Table A–10. Mastering — $PARAM_GROUP[group].$MASTER_POS

[Function]	Store jig (fixture) position for jig (fixture) mastering
[Description]	Jig (fixture) position for jig (fixture) mastering is stored. Mastering pulse count is calculated from this data.

A.2.4 Quick Mastering

$DMR_GRP[group]. $REF_DONE FALSE

BOOLEAN RW TRUE / FALSE

Table A–11. Quick Mastering — $DMR_GRP[group].$REF_DONE

[Function]	Indicates if setting of the reference point for quick mastering is completed.
[Description]	When the reference point of quick mastering is set, the Pulsecoder count and coordinate values of the reference position are stored.
[Setting]	On the positioning screen [6 SYSTEM, Master/Cal]

$DMR_GRP[group]. $REF_COUNT[1] 0

$DMR_GRP[group]. $REF_COUNT[2] 0

$DMR_GRP[group]. $REF_COUNT[3] 0

$DMR_GRP[group]. $REF_COUNT[4] 0

$DMR_GRP[group]. $REF_COUNT[5] 0

$DMR_GRP[group]. $REF_COUNT[6] 0

$DMR_GRP[group]. $REF_COUNT[7] 0

$DMR_GRP[group]. $REF_COUNT[8] 0

$DMR_GRP[group]. $REF_COUNT[9] 0

INTEGER RW 0 to 100000000 (pulse)

Table A–12. Quick Mastering — $DMR_GRP.$REF_COUNT[group].$REF_COUNT

[Function]	Store reference point mastering count
[Description]	Store the count of the Pulsecoder when the robot is positioned at the reference point.

$DMR_GRP[group]. $REF_POS[1] 0

$DMR_GRP[group]. $REF_POS[2] 0

$DMR_GRP[group]. $REF_POS[3] 0

$DMR_GRP[group]. $REF_POS[4] 0

$DMR_GRP[group]. $REF_POS[5] 0

$DMR_GRP[group]. $REF_POS[6] 0

$DMR_GRP[group]. $REF_POS[7] 0

$DMR_GRP[group]. $REF_POS[8] 0

$DMR_GRP[group]. $REF_POS[9] 0

REAL RW -100000 to 100000 (rad)

Table A–13. Quick Mastering — $DMR_GRP.$REF_POS[group].$REF_POS

[Function]	Store reference point to be set during quick mastering
[Description]	Store the reference point to be set during quick mastering.

A.2.5 Calibration

$MOR_GRP[group]. $CAL_DONE TRUE

BOOLEAN RW TRUE / FALSE

Table A–14. Calibration — $MOR_GRP.$CAL_DONE[group]

[Function]	Indicates if calibration is completed.
[Description]	To check the current position of the robot, the count of the Pulsecoder issued and the current position is calculated using mastering count. This check is usually performed when the power is turned on.
[Setting]	On the calibration screen [6 SYSTEM, Master/Cal]

A.2.6 Specifying Coordinate Systems

$MNUFRAMENUM[group] 0

BYTE RW 0 to 9

Table A–15. Specifying Coordinate Systems — $MNUFRAMENUM[group]

[Function]	Specifies user coordinate system number
[Description]	Specifies the number of the user coordinate system currently used. 0: World coordinate system 1 to 9: User coordinate system
[Setting]	On the user coordinate system setting screen [6 SETUP, Frames, User Frame]

$MNUFRAME[group, 1] XYZWPR

$MNUFRAME[group, 2] XYZWPR

$MNUFRAME[group, 3] XYZWPR

$MNUFRAME[group, 4] XYZWPR

$MNUFRAME[group, 5] XYZWPR

$MNUFRAME[group, 6] XYZWPR

$MNUFRAME[group, 7] XYZWPR

$MNUFRAME[group, 8] XYZWPR

$MNUFRAME[group, 9] XYZWPR

POSITION RW XYZWPR

Table A–16. Specifying Coordinate Systems — $MNUFRAME[group]

[Function]	Specifies user coordinates system
[Description]	Specifies the Cartesian coordinates in the user coordinate system. Up to nine user coordinate systems can be registered.

$MNUTOOLNUM[group] 0

BYTE RW 0 to 10

Table A–17. Specifying Coordinate Systems — $MNUTOOLNUM

[Function]	Specifies tool coordinate system number
[Description]	Specifies the number of the tool coordinate system currently used. 0: Mechanical interface coordinate system 1 to 10: Tool coordinate system
[Setting]	On the tool coordinate system setting screen [6 SETUP, Frames, Tool Frame]

$MNUTOOL[group, 1] XYZWPR

$MNUTOOL[group, 2] XYZWPR

$MNUTOOL[group, 3] XYZWPR

$MNUTOOL[group, 4] XYZWPR

$MNUTOOL[group, 5] XYZWPR

$MNUTOOL[group, 6] XYZWPR

$MNUTOOL[group, 7] XYZWPR

$MNUTOOL[group, 8] XYZWPR

$MNUTOOL[group, 9] XYZWPR

$MNUTOOL[group, 10] XYZWPR

POSITION RW XYZWPR

Table A–18. Specifying Coordinate Systems — $MNUTOOL[group]

[Function]	Specifies the tool coordinate system
[Description]	Specify the Cartesian coordinates in the tool coordinate system. Ten tool coordinate systems can be registered.

$JOG_GROUP[group]. $JOG_FRAME XYZWPR

POSITION RW XYZWPR

Table A–19. Specifying Coordinate Systems — $JOG_GROUP[group].$JOG_FRAME

[Function]	Specifies the jog coordinate system
[Description]	Specifies the Cartesian coordinates in the jog coordinate system.
[Setting]	On the jog coordinate system setting screen [6 SETUP, Frames, Jog Frame]

A.2.7 Setting Motors

$SCR_GRP[group]. $AXISORDER[1] 1

$SCR_GRP[group]. $AXISORDER[2] 2

$SCR_GRP[group]. $AXISORDER[3] 3

$SCR_GRP[group]. $AXISORDER[4] 4

$SCR_GRP[group]. $AXISORDER[5] 5

$SCR_GRP[group]. $AXISORDER[6] 6

$SCR_GRP[group]. $AXISORDER[7] 0

$SCR_GRP[group]. $AXISORDER[8] 0

$SCR_GRP[group]. $AXISORDER[9] 0

BYTE RW 0 to 16

Table A–20. Setting Motors — $SCR_GRP[group].$AXISORDER

[Function]	Specify axis order
[Description]	Specifies the order of axes by assigning the physical number of a servo motor controlled by the servo amplifier (servo register) to the logical number of a robot joint axis specified in software (Jx-axis). For instance, when $AXISORDER[1] = 2, servo motor 2 is assigned to the J1-axis. When $AXISORDER[1] = 0, no servo motor is assigned as the J1-axis.

$SCR_GRP[group]. $ROTARY_AXS[1] *

$SCR_GRP[group]. $ROTARY_AXS[2] *

$SCR_GRP[group]. $ROTARY_AXS[3] *

$SCR_GRP[group]. $ROTARY_AXS[4] *

$SCR_GRP[group]. $ROTARY_AXS[5] *

$SCR_GRP[group]. $ROTARY_AXS[6] *

$SCR_GRP[group]. $ROTARY_AXS[7] *

$SCR_GRP[group]. $ROTARY_AXS[8] *

$SCR_GRP[group]. $ROTARY_AXS[9] *

BOOLEAN RO TRUE / FALSE

Table A–21. Setting Motors — $SCR_GRP[group].$ROTARY_AXS

[Function]	Specify axis type
[Description]	Specifies whether joint axes of the robot are rotational or linear. TRUE: Rotational FALSE: Linear

$PARAM_GROUP[group]. $MOSIGN[1] *

$PARAM_GROUP[group]. $MOSIGN[2] *

$PARAM_GROUP[group]. $MOSIGN[3] *

$PARAM_GROUP[group]. $MOSIGN[4] *

$PARAM_GROUP[group]. $MOSIGN[5] *

$PARAM_GROUP[group]. $MOSIGN[6] *

$PARAM_GROUP[group]. $MOSIGN[7] *

$PARAM_GROUP[group]. $MOSIGN[8] *

$PARAM_GROUP[group]. $MOSIGN[9] *

BOOLEAN RW PU TRUE / FALSE

Table A–22. Setting Motors — $PARAM_GROUP[group].$MOSIGN

[Function]	Specify direction of rotation around axes
[Description]	Specify whether the robot moves in the positive or negative direction when the motor rotates positively for each axis. TRUE: The robot moves in a positive direction when the motor rotates positively. FALSE: The robot moves in a negative direction when the motor rotates positively.

$PARAM_GROUP[group]. $ENCSCALES[1] *

$PARAM_GROUP[group]. $ENCSCALES[2] *

$PARAM_GROUP[group]. $ENCSCALES[3] *

$PARAM_GROUP[group]. $ENCSCALES[4] *

$PARAM_GROUP[group]. $ENCSCALES[5] *

$PARAM_GROUP[group]. $ENCSCALES[6] *

$PARAM_GROUP[group]. $ENCSCALES[7] *

$PARAM_GROUP[group]. $ENCSCALES[8] *

$PARAM_GROUP[group]. $ENCSCALES[9] *

REAL RW PU -10000000000 to 10000000000 (pulse/deg, pulse/mm)

Table A–23. Setting Motors — $PARAM_GROUP[group].$ENCSCALES

[Function]	Specify unit of Pulsecoder count
[Description]	Specify how many pulses are required for the Pulsecoder when the robot moves around a joint axis one degree or the robot moves along a joint axis 1 mm. Rotation axis: $ENCSCALES = 2E19 x deceleration ratio/360

$PARAM_GROUP[group]. $MOT_SPD_LIM[1] *

$PARAM_GROUP[group]. $MOT_SPD_LIM[2] *

$PARAM_GROUP[group]. $MOT_SPD_LIM[3] *

$PARAM_GROUP[group]. $MOT_SPD_LIM[4] *

$PARAM_GROUP[group]. $MOT_SPD_LIM[5] *

$PARAM_GROUP[group]. $MOT_SPD_LIM[6] *

$PARAM_GROUP[group]. $MOT_SPD_LIM[7] *

$PARAM_GROUP[group]. $MOT_SPD_LIM[8] *

$PARAM_GROUP[group]. $MOT_SPD_LIM[9] *

INTEGER RW PU 0 to 100000 (rpm)

Table A–24. Setting Motors — $PARAM_GROUP[group].$MOT_SPD_LIM

[Function]	Specify maximum motor speed
[Description]	Specifies the maximum speed of each servo motor for the robot for each axis. When the robot moves around or along a certain axis at a speed exceeding the maximum speed, a warning is issued. Then, the robot decelerates and moves at a speed not exceeding the maximum speed. In this case, the robot may not trace the specified path.

A.2.8 Override

$SHIFTOV_ENB 0

ULONG RW 0 / 1

Table A–25. Override — $SHIFTOV_ENB

[Function]	Enables or disables shift override
[Description]	The shift override function changes the feed rate override in five steps. To change the feed rate override, press and hold down the SHIFT key, then press the override key as many times as necessary to select the desired override. 1: Enables shift override. 0: Disables shift override. Press and hold down [SHIFT] key, then press the override key: The feed rate override changes in the order: VFINE —> FINE —> 5% —> 50% —> 100%.

$MCR. $PROGOVERRIDE 100

INTEGER RW 0 to 100 (%)

Table A–26. Override — $MCR.$PROGOVERRIDE

[Function]	Specifies program override
[Description]	Specifies the percentage of the robot feed rate while the program is being played back.

$SCR_GRP . $JOGLIM 12

INTEGER RO 0 to 100 (%)

Table A–27. Override — $SCR_GRP.$JOGLIM

[Function]	Maximum speed scale for coordinate jogging
[Description]	Percentage of the maximum speed when jogging the robot in the x, y, or z directions using XYZ or TOOL frame. The maximum speed of linear motion is specified by $PARAM_GROUP[group].$SPEEDLIM.

$SCR_GRP . $JOGLIMROT 12

INTEGER RO 0 to 100 (%)

Table A–28. Override — $SCR_GRP.$JOGLIMROT

[Function]	Maximum speed scale for orientation jogging
[Description]	Percentage of the maximum speed when jogging the robot about the x, y, or z axes using XYZ or TOOL frame.

$SCR_GRP[group]. $JOGLIM_JNT[1] *

$SCR_GRP[group]. $JOGLIM_JNT[2] *

$SCR_GRP[group]. $JOGLIM_JNT[3] *

$SCR_GRP[group]. $JOGLIM_JNT[4] *

$SCR_GRP[group]. $JOGLIM_JNT[5] *

$SCR_GRP[group]. $JOGLIM_JNT[6] *

$SCR_GRP[group]. $JOGLIM_JNT[7] *

$SCR_GRP[group]. $JOGLIM_JNT[8] *

$SCR_GRP[group]. $JOGLIM_JNT[9] *

INTEGER RO 0 to 100 (%)

Table A–29. Override — $SCR_GRP[group].$JOGLIM_JNT

[Function]	Specify joint jog override
[Description]	The joint jog override function specifies the percentage of the robot feed rate for each axis during jog feed. Specify a low jog override because it is generally unnecessary to move the robot at high speed, and because it is always prudent to avoid danger.

$SCR. $COLDOVRD 10

INTEGER RO 0 to 100 (%)

Table A–30. Override — $SCR.$COLDOVRD

[Function]	Specifies maximum feed rate override after a cold start
[Description]	The feed rate override is set to this value after a cold start.

$SCR. $COORDOVRD 10

INTEGER RW 0 to 100 (%)

Table A–31. Override — $SCR.$COORDOVRD

[Function]	Specifies maximum feed rate override when the manual-feed coordinate system is changed
[Description]	The feed rate override is set to this value or less when the manual-feed coordinate system is changed.

$SCR. $TPENBLOVRD 10

INTEGER RO 0 to 100 (%)

Table A–32. Override — $SCR.$TPENBLOVRD

[Function]	Specifies the maximum feed rate override when the teach pendant is enabled
[Description]	The feed rate override is set to this value when the teach pendant is enabled.

$SCR. $JOGOVLIM 100

INTEGER RO 0 to 100 (%)

Table A–33. Override — $SCR.$JOGOVLIM

[Function]	Specifies the maximum feed rate override during jog feed
[Description]	The feed rate override is set to this value or less during jog feed.

$SCR. $RUNOVLIM 100

INTEGER RW 0 to 100 (%)

Table A–34. Override — $SCR.$RUNOVLIM

[Function]	Specifies the maximum feed rate override when the program is executed
[Description]	The feed rate override is set to this value or less when the program is executed.

$SCR. $FENCEOVRD 10

INTEGER RO 0 to 100 (%)

Table A–35. Override — $SCR.$FENCEOVRD

[Function]	Maximum feed rate override when the safety fence is open
[Description]	When the safety fence is opened (*SFSPD input is turned off), the feed rate override is set to this value or below.

$SCR. $SFJOGOVLIM 50

INTEGER RO 0 to 100 (%)

Table A–36. Override — $SCR.$SFJOGOVLIM

[Function]	Maximum feed rate override of jog feed when the safety fence is open
[Description]	If jog feed is performed while the safety fence is open, the feed rate override is set to this value or below.

$SCR. $SFRUNOVLIM 30

INTEGER RO 0 to 100 (%)

Table A–37. Override — $SCR.$SFRUNOVLIM

[Function]	Maximum feed rate override of program execution while the safety fence is open
[Description]	When a program is executed with the safety fence open (*SFSPD input set off), the feed rate override is set to this value or below.

$SCR. $RECOV_OVRD FALSE

BOOLEAN RW TRUE/FALSE

Table A–38. Override — $SCR.$RECOV_OVRD

[Function]	Function to restore feed rate override when the safety fence is closed
[Description]	When the safety fence is closed (*SFSPD input set on), the previous feed rate override is restored. Then, automatic operation can be started immediately. This function is enabled when the following conditions are satisfied: 1. $SCR.$RECOV_OVRD is set to TRUE. 2. The system is in the remote state. 3. The feed rate override is not changed while the safety fence is open. If the safety fence is closed while the above conditions are not satisfied, the previous override cannot be restored.
[Setting]	General item setting screen [6 SETUP, General] Jog feed rate(joint feed) (deg/sec, mm/sec) = $\text{Maximum joint feed rate} \times \frac{\text{Each axis jog override}}{100} \times \frac{\text{Feedrate override}}{100}$ Jog feed rate(linear feed) (mm/sec) = $\text{Maximum linear feed rate} \times \frac{\text{Jog override}}{100} \times \frac{\text{Feedrate override}}{100}$ Jog feed rate(Circular feed) (deg/sec) = $\text{Maximum circular feed rate} \times \frac{\text{Orientation Jog override}}{100} \times \frac{\text{Feedrate override}}{100}$ Each axis jog override: $SCR_GRP[g]. $JOGLIM_JNT[i] (%) Jog override: $SCR. $JOGLIM (%) Orientation jog override: $SCR. $JOGLIMROT (%) Note: g is group number. i is axis number. Operating speed (joint control motion) (deg/sec, mm/sec) = $\text{Maximum joint feed rate} \times \frac{\text{Programmed feedrate}}{100} \times \frac{\text{Feedrate override}}{100}$ Operating speed (motion under path control) (mm/sec) = $\text{Programmed feed rate} \times \frac{\text{Feedrate override}}{100}$ Operating speed (motion under attitude control) (deg/sec) = $\text{Programmed feed rate} \times \frac{\text{Feedrate override}}{100}$

A.2.9 Payload Specification

Note Use Payload Setting screen (Motion Performance Screen) to set values of payload weight, gravity center position and inertia.

$PARAM_GROUP [group]. $AXISINERTIA[1] *

$PARAM_GROUP [group]. $AXISINERTIA[2] *

$PARAM_GROUP [group]. $AXISINERTIA[3] *

$PARAM_GROUP [group]. $AXISINERTIA[4] *

$PARAM_GROUP [group]. $AXISINERTIA[5] *

$PARAM_GROUP [group]. $AXISINERTIA[6] *

$PARAM_GROUP [group]. $AXISINERTIA[7] *

$PARAM_GROUP [group]. $AXISINERTIA[8] *

$PARAM_GROUP [group]. $AXISINERTIA[9] *

SHORT RW PU 0 to 32767 (kgf · cm · sec^2)

Table A–39. Payload Specification — $PARAM_GROUP[group].$AXISINERTIA

[Function]	Payload Inertia
[Description]	Indicates the value of Inertia about each axis resulting from the payload. The values for the 1st to 3rd axes are calculated automatically; therefore, they need not be specified. (Set a value for each of the 4th, 5th, and 6th axes.) The inertia for each axis is calculated using the following expression: $\$AXISINERTIA[i] = \dfrac{payload \times (l\_max[i])^2}{g} \quad (kgf \cdot cm \cdot sec^2)$ Payload : Payload [kgf] l_max[i] : Maximum distance from the rotation center of the axis (axis i) to the mass center of the load on the robot [cm] For the 4th and 5th axes, the distance may vary depending on the angle of the other axes. In such a case, set the maximum distance that can be achieved. g : Gravity acceleration (= 980 [cm/sec^2])

Table A–39. Payload Specification — $PARAM_GROUP[group].$AXISINERTIA (Cont'd)

[NOTE 1] When specifying or changing this variable, refer to the explanation of $PARAM_GROUP[].$AXIS_IM_SCL, below.
[NOTE 2] If $PARAM_GROUP[group].$SV_DMY_LNK[4] is TRUE, these system variables are automatically updated according to the present payload setting.

$PARAM_GROUP [group]. $AXISMOMENT[1] *

$PARAM_GROUP [group]. $AXISMOMENT[2] *

$PARAM_GROUP [group]. $AXISMOMENT[3] *

$PARAM_GROUP [group]. $AXISMOMENT[4] *

$PARAM_GROUP [group]. $AXISMOMENT[5] *

$PARAM_GROUP [group]. $AXISMOMENT[6] *

$PARAM_GROUP [group]. $AXISMOMENT[7] *

$PARAM_GROUP [group]. $AXISMOMENT[8] *

$PARAM_GROUP [group]. $AXISMOMENT[9] *

SHORT RW PU 0 to 32767 (kgf · m)

Table A–40. Payload Specification — $PARAM_GROUP[group].$AXISMOMENT

[Function]	Axis moment
[D [Function] escription]	Indicates the value of Moment about each axis resulting from the payload. The values for the 1st to 3rd axes are calculated automatically; therefore, they need not be specified. (Set a value for each of the each of 4th, 5th, and 6th axes.) The moment value for each axis is calculated using the following expression: $AXISMOMENT[i] = payload × l_max[i] (kgf·m) Payload : Payload [kgf] l_max[i] : Maximum distance from the rotation center of the axis (axis i) to the mass center of the load on the robot [m]. For the 4th and 5th axes, the distance may vary depending on the angle of the other axes. In such a case, set the maximum distance that can be achieved.
[NOTE 1] When specifying or changing this variable, refer to the explanation of $PARAM_GROUP[].$AXIS_IM_SCL, below.	
[NOTE 2] If $PARAM_GROUP[group].$SV_DMY_LNK[4] is TRUE, these system variables are automatically updated according to the present payload setting.	

$PARAM_GROUP [group] . $AXIS_IM_SCL 1

SHORT RW PU 0 to 32767

Table A–41. Payload Specification — $PARAM_GROUP[group].$AXIS_IM_SCL

<table>
<tr><td>[Function]</td><td>Inertia and moment value adjustment scale</td></tr>
<tr><td>[Description]</td><td>This scale is used to set up a number in decimal places for the inertia and moment values of each axis stated above.</td></tr>
<tr><td colspan="2">[NOTE] It is usually unnecessary to re-set this variable. Actually, the following inertia and moment values are used.</td></tr>
<tr><td></td><td>$$(\text{Inertia value}) = \frac{\text{\$PARAM\_GROUP[group].\$AXISINERTIA[i]}}{\text{\$PARAM\_GROUP[group].\$AXIS\_IM\_SCL}}$$
$$(\text{Moment value}) = \frac{\text{\$PARAM\_GROUP[group].\$AXISMOMETN[i]}}{\text{\$PARAM\_GROUP[group].\$AXIS\_IM\_SCL}}$$
It is therefore necessary to assign $AXISINERTIA[i] and $AXISMOMENT[i] with values that match the setting of this variable.

To enter the value "1.23," for example, as the inertia value for the fourth axis of the robot:
• Set up $PARAM_GROUP[group].$AXIS_IM_SCL = 100
• Set up $PARAM_GROUP[group].$AXISINERTIA[4] = 123
• Change these inertia and moment values for other axes according to the value of $AXIS_IM_SCL.</td></tr>
</table>

A.2.10 Executing a Program

$DEFPULSE 4

SHORT RW 0 to 255 (100 msec)

Table A–42. Executing a Program — $DEFPULSE

[Function]	Specifies the standard DO output pulse width
[Description]	This value is used when the pulse width is not specified for the output of a DO signal pulse.

A.2.11 Automatic Operation

$RMT_MASTER 0

INTEGER RW 0 to 3

Table A–43. Automatic Operation — $RMT_MASTER

[Function]	Specifies which remote unit is used
[Description]	Specifies which remote unit is used. The specified remote unit has the right to start the robot. 0: Peripheral unit (remote controller) 1: CRT/keyboard 2: Host computer 3: No remote unit

A.2.12 Deleting the Warning History

$ER_NOHIS 0

BYTE RW 0 / 3

Table A–44. Deleting the Warning History — $ER_NOHIS

[Function]	Warning history delete function
[Description]	WARN alarms, NONE alarms and resets can be deleted from the alarm history. 0: Disables the function. (All alarms and resets are recorded in the history.) 1: Does not record WARN and NONE alarms in the history. 2: Does not record resets. 3: Does not record resets, WARN alarms, and NONE alarms.

A.2.13 Disabling Alarm Output

$ER_NO_ALM. $NOALMENBL 0

BYTE RW 0 / 1

Table A–45. Disabling Alarm Output — $ER_NO_ALM.$NOALMENBL

[Function]	Enables the no-alarm output function
[Description]	When this function is enabled, the LEDs on the teach pendant and the machine operator's panel corresponding to the alarms specified with system variable $NOALM_NUM do not light. In addition, the peripheral I/ O alarm signal (FAULT) is not output.

$ER_NO_ALM. $NOALM_NUM 5

BYTE RW 0 to 10

Table A–46. Disabling Alarm Output — $ER_NO_ALM.$NOALM_NUM

[Function]	Specifies the number of alarms not output
[Description]	Specifies the number of alarms that are not output.

$ER_NO_ALM. $ER_CODE1 11001

$ER_NO_ALM. $ER_CODE2 11002

$ER_NO_ALM. $ER_CODE3 11003

$ER_NO_ALM. $ER_CODE4 11007

$ER_NO_ALM. $ER_CODE5 11037

$ER_NO_ALM. $ER_CODE6 0

$ER_NO_ALM. $ER_CODE7 0

$ER_NO_ALM. $ER_CODE8 0

$ER_NO_ALM. $ER_CODE9 0

$ER_NO_ALM. $ER_CODE10 0

INTEGER RW 0 to 100000

Table A–47. Disabling Alarm Output — $ER_NO_ALM.$ER_CODE

[Function]	Specify the alarms not output
[Description]	Specify the alarms that are not output. Setting : $\underbrace{1\ 1}_{\text{Alarm ID}}\ \underbrace{0\ 0\ 2}_{\text{Alarm number}}$ (Meaning: SERVO-002 alarm)

A.2.14 User Alarm

$UALRM_SEV[ ] 6

BYTE RW 0 to 255

Table A–48. User Alarm — $UALRM_SEV

[Function]	User alarm severity
[Description]	Sets the user alarm severity. $UALRM_SEV[i] corresponds to the severity of user alarm[i]. 0 WARN 6 STOP.L 38 STOP.G 11 ABORT.L 43 ABORT.G The initial severity for each user alarm is 6 (STOP.L).

A.2.15 Jogging

$JOG_GROUP. $FINE_DIST 0.5

REAL RW 0.0 to 1.0 (mm)

Table A–49. Jogging — $JOG_GROUP.$FINE_DIST

[Function]	Move distance for linear step jogging
[Description]	Specify an amount of travel in low-speed linear step feed by Cartesian/tool manual feed. The amount of travel in very low speed step feed is one tenth of the value specified here.

$SCR . $FINE_PCNT 10

INTEGER RO 1 to 100 %

Table A–50. Jogging — $SCR.$FINE_PCNT

[Function]	Move distance for joint or orientation step jogging
[Description]	Specify an amount of travel for step feed in attitude rotation by axial manual feed or Cartesian/tool manual feed. Specify manual feed with a percentage and an override of 1%.

A.2.16 I/O Setting

$OPWORK . $UOP_DISABLE *

BYTE RW 0 / 1

Table A–51. I/O Setting — $OPWORK.$UOP_DISABLE

[Function]	Enable/disable UOP I/O
[Description]	Specify whether the peripheral equipment input signal is enabled or disabled. If the peripheral equipment input signal is enabled when the robot is operated without any peripheral equipment connected, an alarm cannot be cleared. By disabling the signal with this setting, the alarm can be cleared. When any peripheral equipment is connected, set this variable to 0 before using that equipment.

$SCR . $RESETINVERT FALSE

BOOLEAN RW TRUE / FALSE

Table A–52. I/O Setting — $SCR.$RESETINVERT

[Function]	FAULT_RESET input signal detection.
[Description]	When you set this value to "TRUE", an error is reset by rising edge of FAULT_RESET input signal. If "FALSE" is set, an error is reset by falling edge is detected. TRUE: Check rising edge of reset input signal. FALSE: Check falling edge of reset input signal.

$PARAM_GROUP . $PPABN_ENBL FALSE

BOOLEAN RW TRUE / FALSE

Table A–53. I/O Setting — $PARAM_GROUP.$PPABN_ENBL

[Function]	Enable/disable pressure abnormal *PPABN input
[Description]	Specifies if pressure abnormal signal is detected or not. If you want to use *PPABN input, you should set this variable to TRUE. TRUE: Enable FALSE: Disable

$PARAM_GROUP. $BELT_ENBLE FALSE

BOOLEAN RW TRUE / FALSE

Table A–54. I/O Setting — $PARAM_GROUP.$BELT_ENBLE

[Function]	Belt rupture signal enabled/disabled
[Description]	Specify whether the belt rupture signal (RI[7]) is detected. TRUE: Belt rupture signal enabled FALSE: Belt rupture signal disabled

A.2.17 Software Version

$ODRDSP_ENB 0

ULONG RW 1 / 0

Table A–55. Software Version — $ODRDSP_ENB

[Function]	Display of an order file
[Description]	An order listing, showing the configuration of the software components installed in the controller can be displayed on the display (order file screen) of the teach pendant.

A.2.18 Soft Float Function

$SFLT_ERRTYP 0

INTEGER RW 1 to 10

Table A–56. Soft Float Function — $SFLT_ERRTYP

[Function]	Flag for specifying the alarm to be generated when time-out occurs during follow-up processing of the soft float function
[Description]	This variable specifies the alarm (a servo alarm or program pause alarm) to be generated if a time-out occurs during follow-up processing of the soft float function. 0: Generates servo alarm "SRVO-111 Softfloat time out". 1: Generates program pause alarm "SRVO-112 Softfloat time out".

$SFLT_DISFUP FALSE

BOOLEAN RW TRUE / FALSE

Table A–57. Soft Float Function — $SFLT_DISFUP

[Function]	Specifies whether to perform follow-up processing at the start of each motion instruction.
[Description]	Specify whether to perform follow-up processing of the soft float function at the start of each program motion instruction. TRUE: Does not perform follow-up processing at the start of each program motion instruction. FALSE: Performs follow-up processing at the start of each program motion instruction.

A.2.19 Saving Files

$FILE_APPBCK

Table A–58. Saving Files — $FILE_APPBCK

[Description]	On the file screen, displays the name of a file to be saved as Application.

$FILE_SYSBCK

Table A–59. Saving Files — $FILE_SYSBCK

[Description]	On the file screen, displays the name of a file to be saved as System file.

A.2.20 Register Speed Specification Function

$RGSPD_PREXE FALSE

BOOLEAN RO TRUE/FALSE

Table A–60. Register Speed Specification Function — $RGSPD_PREXE

[Function]	Advanced register speed read enabled or disabled
[Description]	Specify whether an advanced read of operation statement is performed (enabled) or not (disabled) when the movement speed specified by an operation statement is held in a register. TRUE: Advanced read enabled. FALSE: Advanced read disabled.

Caution

When an advanced register speed read is enabled with the setting indicated above, the timing at which the register value is changed is important. With some timings, a change in the register value may not be reflected in the operation speed, and the register value existing before the change may be applied to the movement. To enable advanced register speed read, some consideration is needed: The value of a register used for the movement speed during program execution should not be changed; An interlock should be provided.

A.2.21 Specifying an Output Signal of the BZAL/BLAL Alarm

$BLAL_OUT.$DO_INDEX 0

INTEGER RW 0 to 256

Table A–61. Specifying an Output Signal of the BZAL/BLAL Alarm —$BLAL_OUT.$DO_INDEX

[Description]	When a non-zero number is specified, DO corresponding to that number is turned on at the occurrence of BZAL/BLAL. DO stays on until the voltage is restored by the replacement backup battery or some other means. (If a program or the I/O screen is used to turn off DO forcibly, DO is turned back on immediately.)

$BLAL_OUT.$BATALM_OR TRUE

INTEGER RW TRUE/FALSE

Table A–62. Specifying an Output Signal of the BZAL/BLAL Alarm —$BLAL_OUT.$BATALM_OR

[Description]	Specifies whether to set BATALM, a dedicated output signal, so that it has also the BZAL/BLAL function.

Caution

In case that the BATALM signal is specified to include BZAL/BLAL of Pulsecoder indicated above, BATALM is output when at least one Pulsecoder of all axis of all motion group detects BZAL/BLAL. However BATALM signal excludes Process axis (Servo torch axis) which has no motion group and Slave axis of Dual drive function.

A.2.22 Setup for Changing Jog Group According to the Motion Group of the Selected Program

$PROGGRP_TGL 0

INTEGER RW 0 to 2147483647

Table A–63. Setup for Changing Jog Group According to the Motion Group of the Selected Program — $PROGGRP_TGL

[Description]	1Bit(1):Setup for changing jog group according to the motion group of selected program 0:Disable (Default) 1:Enable (Jog group is changed step by step according to the motion group of selected program.) 2Bit(2):Setup for enable settings of 1Bit and 3Bit in only case T1 mode when 1Bit or 3Bit is enabled. 0: Settings of 1Bit and 3Bit is enabled regardless of the state of mode switch. 1: Settings of 1Bit and 3Bit is enabled in only T1 mode. 3Bit(4) : Setup for changing jog group automatically when program is selected 0: Disable (Default) 1: Enable (Jog group is changed to motion group of selected program when program is selected in select menu.)

Note n of nBit means low n digit in binary digit. For example, in case 1Bit is 1 and 3Bit is 1 others are 0, “00000101” in binary digit represents “5” in decimal digit. So input “5” to $PROGGRO_TGL.

A.2.23 Default Setting for the Motion Group

$DSBL_GPMSK 0

INTEGER RW 0 to 255

Table A–64. Default Setting for the Motion Group — $DSBL_GPMSK

[Description]	The specified motion group of the program is disable when the program is created. The motion group is specified as bit (1-8) of this variable. For example, motion groups 1 and 3 are disable in case $DSBL_GPMSK is 5, the default motion group of the program will be setup to [*,1,*,1,*,*, *,*] in case system has 4 group when program is created.

Note n of nBit means low n digit in binary digit. For example, in case 1Bit is 1 and 3Bit is 1 others are 0, “00000101” in binary digit represents “5” in decimal digit. So input “5” to $DSBL_GPMSK.

A.2.24 Servo Information

$SV_INFO [group] . $Q_CURRENT [1] *

$SV_INFO [group] . $Q_CURRENT [2] *

$SV_INFO [group] . $Q_CURRENT [3] *

$SV_INFO [group] . $Q_CURRENT [4] *

$SV_INFO [group] . $Q_CURRENT [5] *

$SV_INFO [group] . $Q_CURRENT [6] *

$SV_INFO [group] . $Q_CURRENT [7] *

$SV_INFO [group] . $Q_CURRENT [8] *

$SV_INFO [group] . $Q_CURRENT [9] *

REAL RW -1000000000.0 1000000000.0 (Ap)

Table A–65. Servo Information — $SV_INFO[group].$Q_CURRENT

[Function]	Q-phase current command.
[Description]	The effective component of current Feedback.

$SV_INFO [group] . $AXIS_POS [1] *

$SV_INFO [group] . $AXIS_POS [2] *

$SV_INFO [group] . $AXIS_POS [3] *

$SV_INFO [group] . $AXIS_POS [4] *

$SV_INFO [group] . $AXIS_POS [5] *

$SV_INFO [group] . $AXIS_POS [6] *

$SV_INFO [group] . $AXIS_POS [7] *

$SV_INFO [group] . $AXIS_POS [8] *

$SV_INFO [group] . $AXIS_POS [9] *

REAL RW -1000000000.0 1000000000.0 (deg or mm)

Table A–66. Servo Information — $SV_INFO[group].$AXIS_POS

[Function]	Axis position.
[Description]	Position of each axis. The unit is [deg] for rotated axis, [mm] for linear axis.

A.2.25 System Timer

$SYSTEM_TIMER[group].$PWR_TOT 0

INTEGER RO 0 to 2147483647

Table A–67. System Timer $SYSTEM_TIMER[group].$PWR_TOT

[Function]	On Power time.
[Description]	Time during which the power to the controller is on (Unit: minute).

$SYSTEM_TIMER[group].$SRV_TOT 0

INTEGER RO 0 to 2147483647

Table A–68. $SYSTEM_TIMER[group].$SRV_TOT

[Function]	On Power time.
[Description]	Time during which the system is ready for operation (servo on) after the release of an alarm (Unit: minute).

$SYSTEM_TIMER[group].$RUN_TOT 0

INTEGER RO 0 to 2147483647

Table A–69. $SYSTEM_TIMER[group].$RUN_TOT

[Function]	Running time
[Description]	Program execution time. The halt period is not included (Unit: minute).

$SYSTEM_TIMER[group].$WIT_TOT 0

INTEGER RO 0 to 2147483647

Table A–70. $SYSTEM_TIMER[group].$WIT_TOT

[Function]	Waiting time.
[Description]	Time required to execute a WAIT instruction (Unit: minute).

$SYSTEM_TIMER[group].$SHM_TOT 0

INTEGER RO 0 to 2147483647

Table A–71. $SYSTEM_TIMER[group].$SHM_TOT

[Function]	Servo hour meter time.
[Description]	Time during which servo brake is released (Unit: minute).

A.2.26 System Ready

$PWRUP_DELAY.$SY_READY FALSE

BOOLEAN RO TRUE/FALSE

Table A–72. System Ready — $PWRUP_DELAY.$SY_READY

[Function]	System ready.
[Description]	FALSE means that system is NOT ready(during Start-up). TRUE means that system is ready.